

Input Tensor
CIFAR-100 Grid (32 × 32 × 3)

**ImageNet Pretrained Feature Backbone
(ResNet-50 Feature Space Architecture)**

- Inherited Cross-Domain Priors
- ~23.5 Million Parameters
- Stable Non-Washout Trajectory

Deep Bottleneck Residual Block Cascade

1x1 Conv (Channel Compression)	
3x3 Conv (Spatial Filter)	
1x1 Conv (Channel Expansion)	

$$H(\mathbf{x}) = F(\mathbf{x}, \{W_i\}) + \mathbf{x}$$

Global Average Pooling (GAP) Layer
(Collapses Spatial Planes → 1 × 1 × 2048)

Custom Re-Engineered Linear Head
Linear Classifier (In: 2048 → Out: 100 Classes)